

Setup - Preparation Before You Begin

Install R, RStudio and Packages

Steps needed to get started

1. Install R
2. Install RStudio Desktop
3. Install R Packages
4. Run a quick test
5. Download Files from Google Drive
6. Create New RStudio Project

1. Install R

R versus RStudio

Note: **R** is the name of the programming language itself and **RStudio** is an integrated development environment (IDE) which is an enhanced interface for better organization, files management and analysis workflows.

Download R from CRAN

The R programming language is maintained and distributed by CRAN [The Comprehensive R Archive Network](#). They provide the installers for the binary distributions of the base system and contributed packages. R is FREE!

Installation Instructions are provided for these operating systems:

- [Windows](#)
 - Choose “base” and I suggest you accept the default installation options.

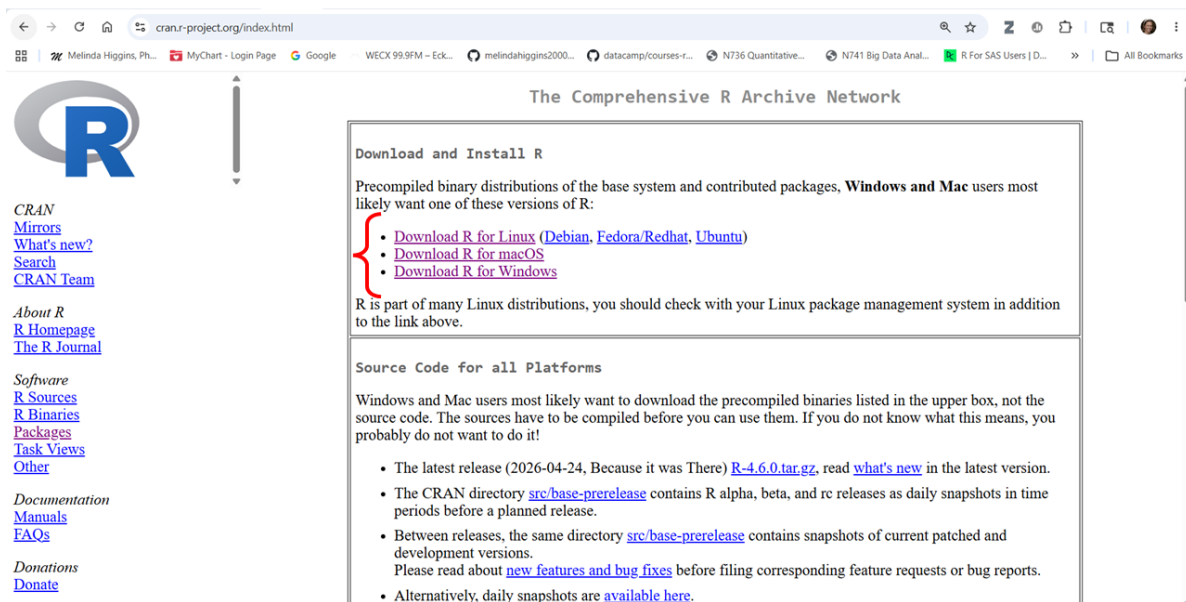
- OPTIONAL - You may also want “RTools” if you think you may want to do build your own R packages in the future. This is NOT needed for this course.

- [Mac](#)

- Please make note of the specific details on which installer you may need for your version of Mac OS (operating system).
- I work on a Windows operating system, so please reach out to your local IT support for help getting the correct version installed for your system.

- [Linux](#)

- Odds are you are probably using either Windows or Mac OS, but if you using Linux, please read carefully the specific details for your version of Linux.



Other Helpful Resources For Installing R

- [Chapter 1 of Statistical Inference via Data Science: A ModernDive into R and the Tidyverse \(Second Edition\)](#) by Chester Ismay, Albert Y. Kim, and Arturo Valdivia.
- [Chapter 3.4 of The Epidemiologist R Handbook](#)

2. Install RStudio Desktop

! Install R First and RStudio Second

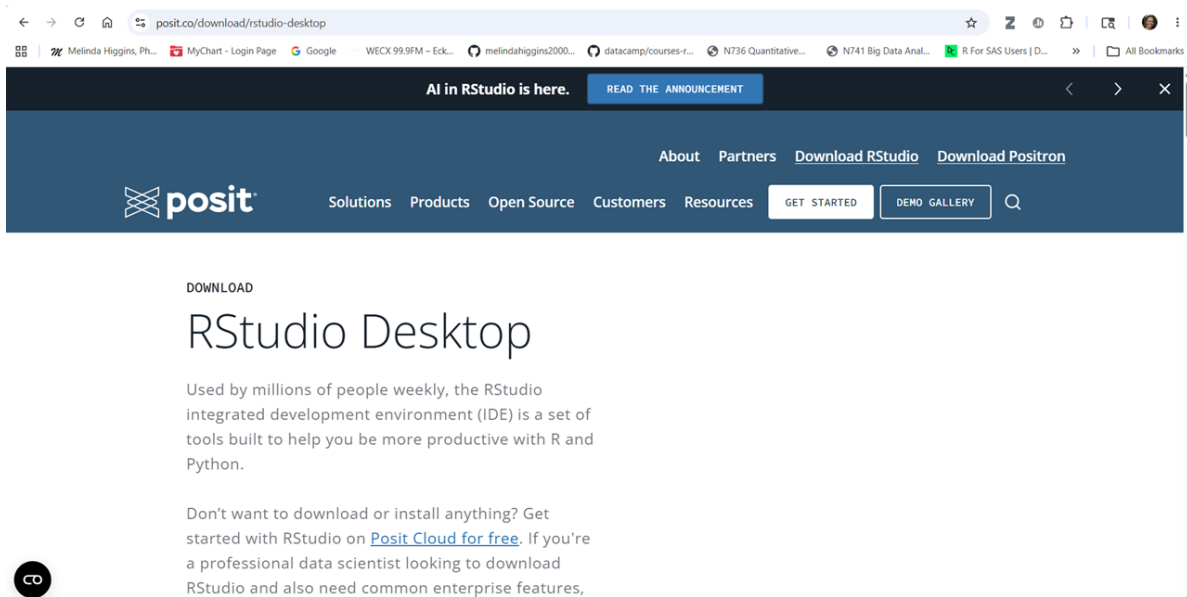
It is important to install the R programming language FIRST. I also encourage you to use the default installation options, so that when you install the RStudio Desktop application, it will “know” where (folder locations) to look for your R installation.

The RStudio IDE (integrated development environment) software is developed and distributed by the [POSIT Company](#), which was formerly known as “RStudio” founded in 2009. In 2022 they changed their name to “Posit”, however, the name of the software application stayed “RStudio”.

For this course, we will be using the RStudio Desktop software. RStudio Desktop is FREE. The “PRO” version does cost money.

You can download RStudio Desktop from <https://posit.co/download/rstudio-desktop/>.

I think the website link above will “autodetect” which operating system you have. Mine defaults to the Windows installer.



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1: Install R

RStudio requires R 3.6.0+. Choose a version of R that matches your computer's operating system.

R is not a Posit product. By clicking on the link below to download and install R, you are leaving the Posit website. Posit disclaims any obligations and all liability with respect to R and the R website.

DOWNLOAD AND INSTALL R

2: Install RStudio

DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS

Size: 319.84 MB | [SHA-256: 8B5DA756](#) | Version: 2026.04.0+526 | Released: 2026-04-22

However, if you have a Mac you should get the installer for the Mac OS. Regardless, you can always choose the operating system and version you need.

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All Installers and Tarballs

RStudio requires a 64-bit operating system.

Linux users may need to import [Posit's public code-signing key](#) prior to installation, depending on the operating system's security policy.

OS	Download	Size	SHA-256
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posit About Partners Download RStudio Download Positron Solutions Products Open Source Customers Resources GET STARTED DEMO GALLERY Q			
OS	Download	Size	SHA-256
Windows OS			
Windows 11	RSTUDIO-2026.04.0-526.EXE ⬇	319.84 MB	8B5DA756
Mac OS			
macOS 14/15/26	RSTUDIO-2026.04.0-526.DMG ⬇	694.56 MB	F49B0016
Linux OS Versions			
Ubuntu 22	RSTUDIO-2026.04.0-526-AMD64.DEB ⬇	238.98 MB	AF4D87C0
Ubuntu 24	RSTUDIO-2026.04.0-526-AMD64.DEB ⬇	238.98 MB	AF4D87C0
Debian 12	RSTUDIO-2026.04.0-526-AMD64.DEB ⬇	238.98 MB	AF4D87C0

And if you keep scrolling down the webpage you can find older versions of RStudio if/as needed as well.

RStudio or Positron?

For this course, I will use the RStudio IDE. However, there is a newer alternative to “RStudio”, known as “Positron” which was released in 2025.

Learn more about [Positron](#). Positron is also an IDE programming environment, but is designed to accommodate multiple programming environments and multiple sessions at the same time, e.g. you can run both R and Python and manage multiple versions at one time. We will NOT cover these details in this course.

Technically, you can also use the RStudio IDE to execute and manage Python programs as well. In fact RStudio can help manage several other programming languages as well.

User Guides

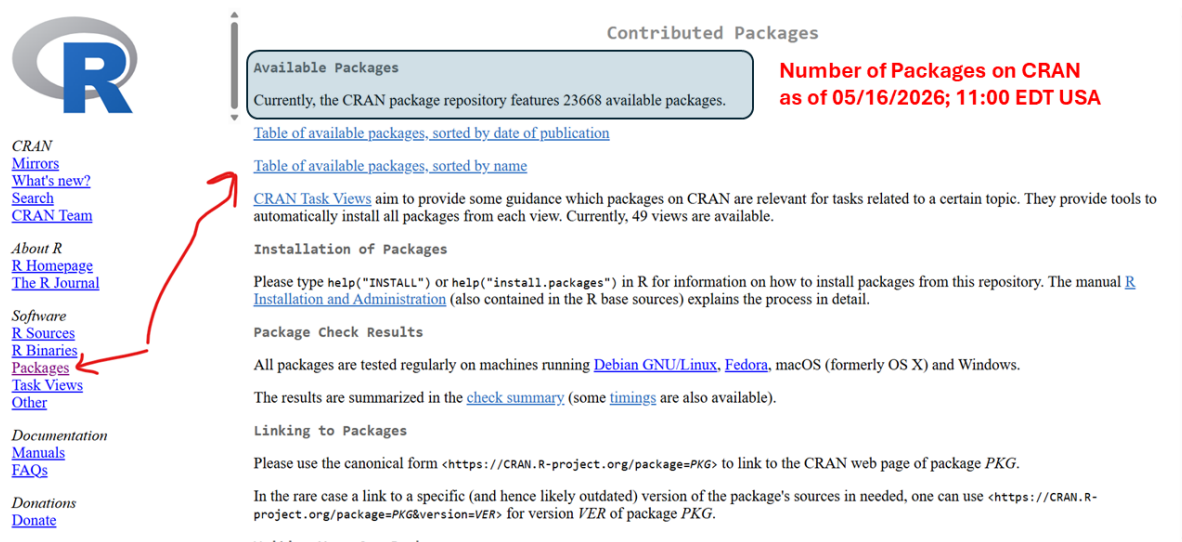
- [RStudio User Guide](#).
- [Positron User Guide](#)

3. Install R Packages Needed for Course

For this course, you will need to have the following packages installed on your computer. All of these packages are available from [CRAN](#).

R Packages on CRAN

As of 05/16/2026 at approximately 11:00 am EDT USA time, there are currently 23668 packages hosted on CRAN. More get added everyday!



Contributed Packages

Available Packages
Currently, the CRAN package repository features 23668 available packages.

**Number of Packages on CRAN
as of 05/16/2026; 11:00 EDT USA**

[Table of available packages, sorted by date of publication](#)
[Table of available packages, sorted by name](#)

[CRAN Task Views](#) aim to provide some guidance which packages on CRAN are relevant for tasks related to a certain topic. They provide tools to automatically install all packages from each view. Currently, 49 views are available.

Installation of Packages
Please type `help("INSTALL")` or `help("install.packages")` in R for information on how to install packages from this repository. The manual [R Installation and Administration](#) (also contained in the R base sources) explains the process in detail.

Package Check Results
All packages are tested regularly on machines running [Debian GNU/Linux](#), [Fedora](#), macOS (formerly OS X) and Windows. The results are summarized in the [check summary](#) (some [timings](#) are also available).

Linking to Packages
Please use the canonical form `<https://CRAN.R-project.org/package=PKG>` to link to the CRAN web page of package *PKG*.
In the rare case a link to a specific (and hence likely outdated) version of the package's sources is needed, one can use `<https://CRAN.R-project.org/package=PKG&version=VER>` for version *VER* of package *PKG*.

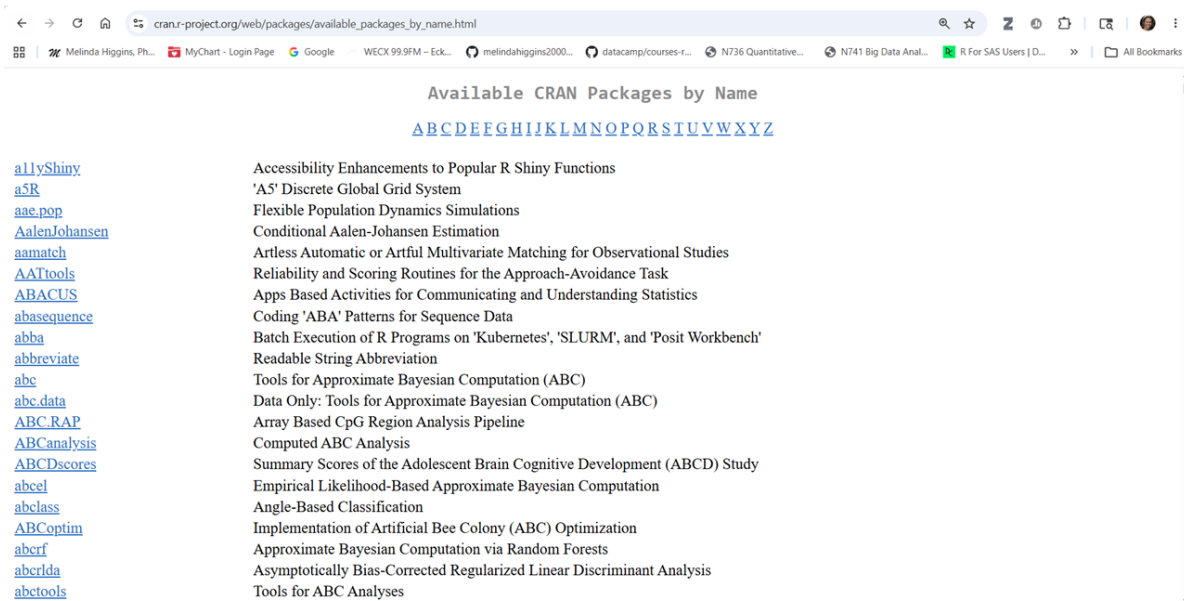
CRAN
[Mirrors](#)
[What's new?](#)
[Search](#)
[CRAN Team](#)

About R
[R Homepage](#)
[The R Journal](#)

Software
[R Sources](#)
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The screenshot shows a web browser window with the URL `cran.r-project.org/web/packages/available_packages_by_name.html`. The page title is "Available CRAN Packages by Name". Below the title is a navigation bar with letters A through Z. The letter 'A' is highlighted. The main content area lists packages starting with 'A'. On the left side of the page, there is a vertical list of package names, each with a blue link. The packages listed are: `a11yShiny`, `a5R`, `aae.pop`, `AalenJohansen`, `aamatch`, `AATtools`, `ABACUS`, `abasequence`, `abba`, `abbreviate`, `abc`, `abc.data`, `ABC.RAP`, `ABCanalysis`, `ABCDscores`, `abcel`, `abclass`, `ABCOptim`, `abcrf`, `abcrlda`, and `abctools`. The main list of packages on the right includes: "Accessibility Enhancements to Popular R Shiny Functions", "'A5' Discrete Global Grid System", "Flexible Population Dynamics Simulations", "Conditional Aalen-Johansen Estimation", "Artless Automatic or Artful Multivariate Matching for Observational Studies", "Reliability and Scoring Routines for the Approach-Avoidance Task", "Apps Based Activities for Communicating and Understanding Statistics", "Coding 'ABA' Patterns for Sequence Data", "Batch Execution of R Programs on 'Kubernetes', 'SLURM', and 'Posit Workbench'", "Readable String Abbreviation", "Tools for Approximate Bayesian Computation (ABC)", "Data Only: Tools for Approximate Bayesian Computation (ABC)", "Array Based CpG Region Analysis Pipeline", "Computed ABC Analysis", "Summary Scores of the Adolescent Brain Cognitive Development (ABCD) Study", "Empirical Likelihood-Based Approximate Bayesian Computation", "Angle-Based Classification", "Implementation of Artificial Bee Colony (ABC) Optimization", "Approximate Bayesian Computation via Random Forests", "Asymptotically Bias-Corrected Regularized Linear Discriminant Analysis", and "Tools for ABC Analyses".

Available CRAN Packages by Name

[A](#) [B](#) [C](#) [D](#) [E](#) [F](#) [G](#) [H](#) [I](#) [J](#) [K](#) [L](#) [M](#) [N](#) [O](#) [P](#) [Q](#) [R](#) [S](#) [T](#) [U](#) [V](#) [W](#) [X](#) [Y](#) [Z](#)

[a11yShiny](#) Accessibility Enhancements to Popular R Shiny Functions
[a5R](#) 'A5' Discrete Global Grid System
[aae.pop](#) Flexible Population Dynamics Simulations
[AalenJohansen](#) Conditional Aalen-Johansen Estimation
[aamatch](#) Artless Automatic or Artful Multivariate Matching for Observational Studies
[AATtools](#) Reliability and Scoring Routines for the Approach-Avoidance Task
[ABACUS](#) Apps Based Activities for Communicating and Understanding Statistics
[abasequence](#) Coding 'ABA' Patterns for Sequence Data
[abba](#) Batch Execution of R Programs on 'Kubernetes', 'SLURM', and 'Posit Workbench'
[abbreviate](#) Readable String Abbreviation
[abc](#) Tools for Approximate Bayesian Computation (ABC)
[abc.data](#) Data Only: Tools for Approximate Bayesian Computation (ABC)
[ABC.RAP](#) Array Based CpG Region Analysis Pipeline
[ABCanalysis](#) Computed ABC Analysis
[ABCDscores](#) Summary Scores of the Adolescent Brain Cognitive Development (ABCD) Study
[abcel](#) Empirical Likelihood-Based Approximate Bayesian Computation
[abclass](#) Angle-Based Classification
[ABCOptim](#) Implementation of Artificial Bee Colony (ABC) Optimization
[abcrf](#) Approximate Bayesian Computation via Random Forests
[abcrlda](#) Asymptotically Bias-Corrected Regularized Linear Discriminant Analysis
[abctools](#) Tools for ABC Analyses

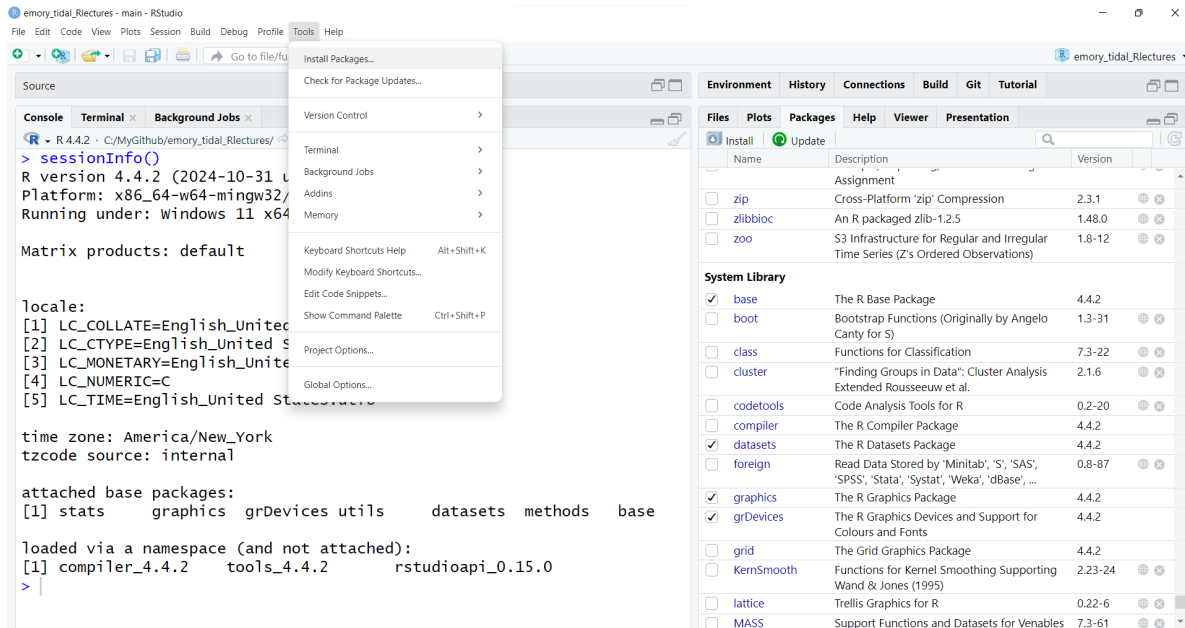
For this course, you will need these 16 packages:

1. `tidyverse` which includes a suite of 9 packages (and a few others as well) that are all installed at the same time:
 - `ggplot2`
 - `dplyr`
 - `tidyr`
 - `readr`
 - `purrr`
 - `tibble`
 - `stringr`
 - `forcats`
 - `lubridate`
 - `haven` - *this package is included in tidyvers*
2. `ggtext`
3. `easystats` which is also a suite of 10 packages:
 - `effectsize`
 - `report`
 - `correlation`
 - `modelbased`
 - `bayestestR`
 - `see`
 - `parameters`
 - `performance`
 - `insight`
 - `datawizard`
4. `gmodels`
5. `gtsummary`
6. `effects`
7. `emmeans`
8. `broom`
9. `rmarkdown`
10. `knitr`
11. `tinytex` - *OPTIONAL* - needed to “knit to PDF”
12. `olsrr`
13. `modelsummary`
14. `janitor`
15. `data.table`
16. `arsenal`

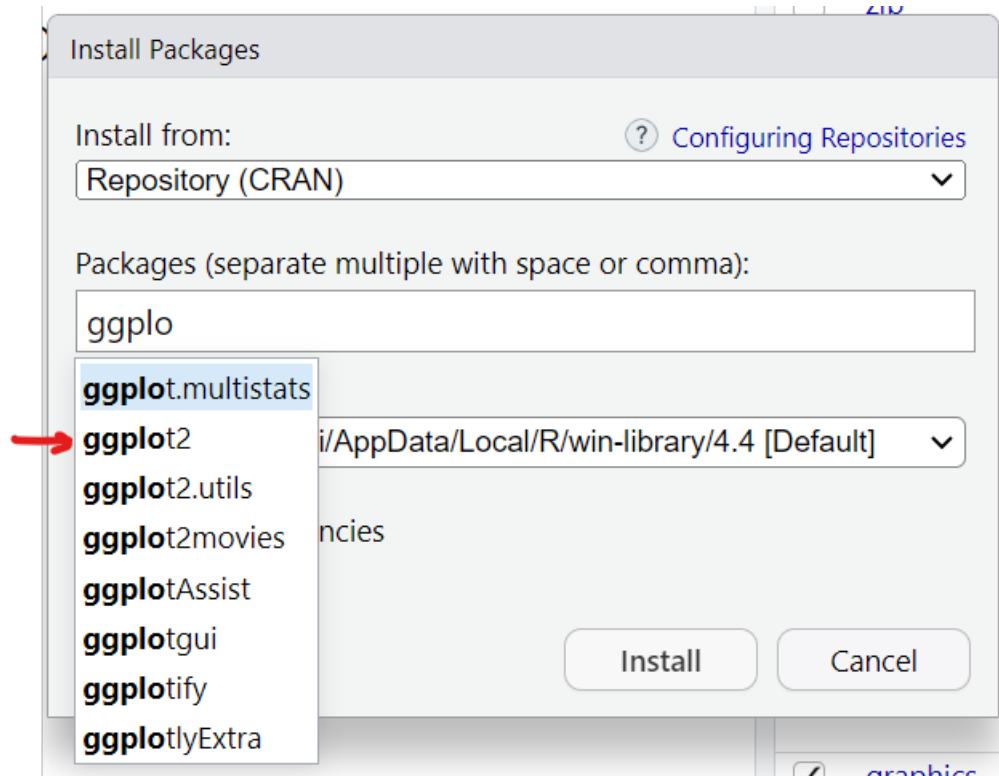
To install these packages, the easiest way is to use the RStudio Menu “Tools/Install” Packages interface:

For example, let’s install a “new” R package, like `ggplot2`.

Go to the RStudio menu “Tools/Install” Packages



This will then open up a window where you can type in the name of the package you want. As soon as we start typing `ggplot2` the menu begins listing all packages with that partial spelling...



You'll notice that there are 3 parts to the installation:

1. Where you want to get the package from (*i.e.*, *which repository* - *more on repositories later in this course*).
2. The name of the package. You can actually type more than one package name at a time separated by commas if you want to install several packages at once.
3. The file location on your computer where the new package is installed - your file location will be different than mine. But this is useful to know in case something goes wrong. **I would suggest keeping the default settings.**

Install Packages

Install from: **1** [? Configuring Repositories](#)

Package Repository (CRAN) ▼

Packages (separate multiple with space or comma): **2**

ggplot2

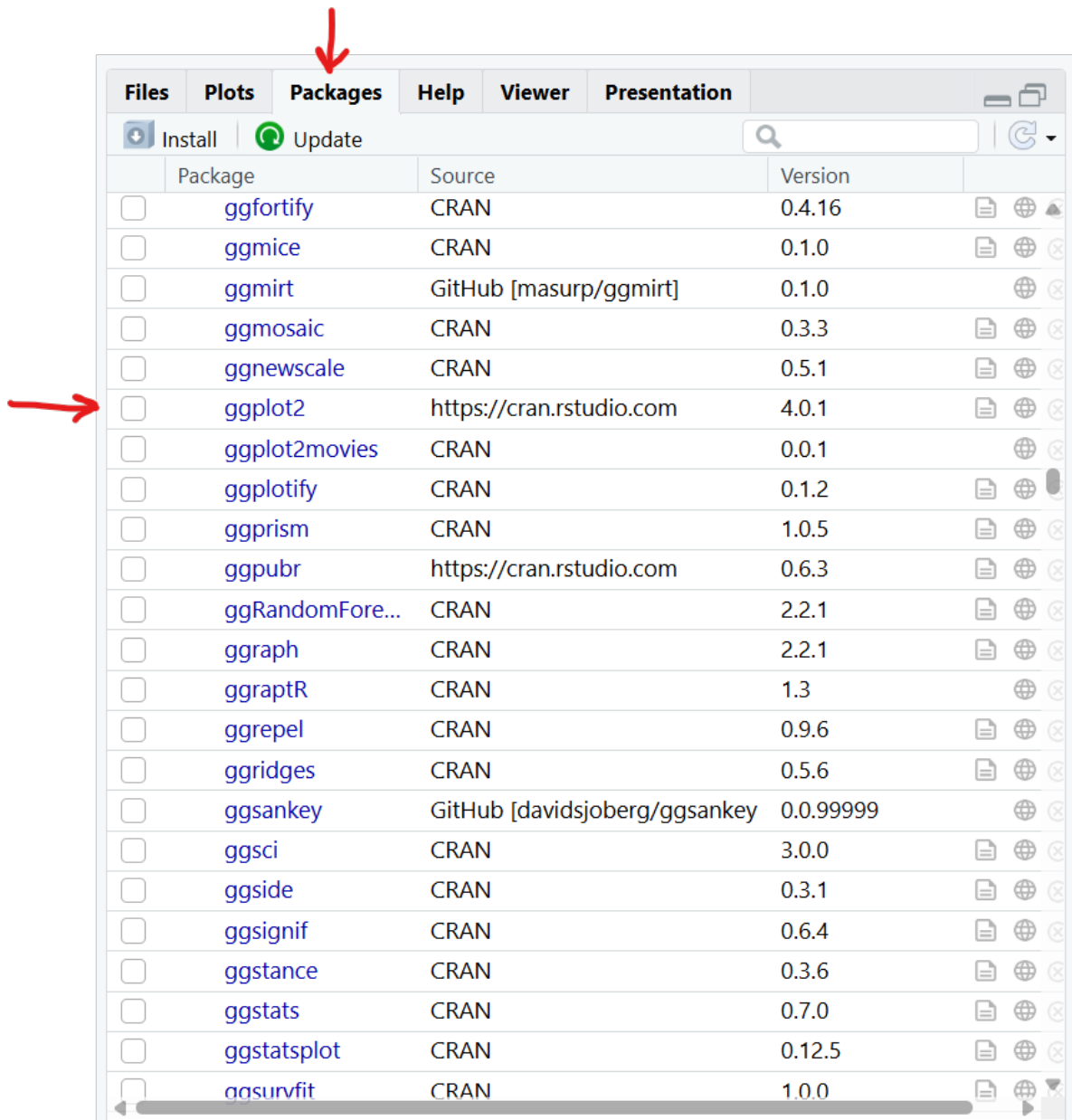
Install to Library: **3**

C:/Users/mkhiggi/AppData/Local/R/win-library/4.6 [Default] ▼

☒ Install dependencies

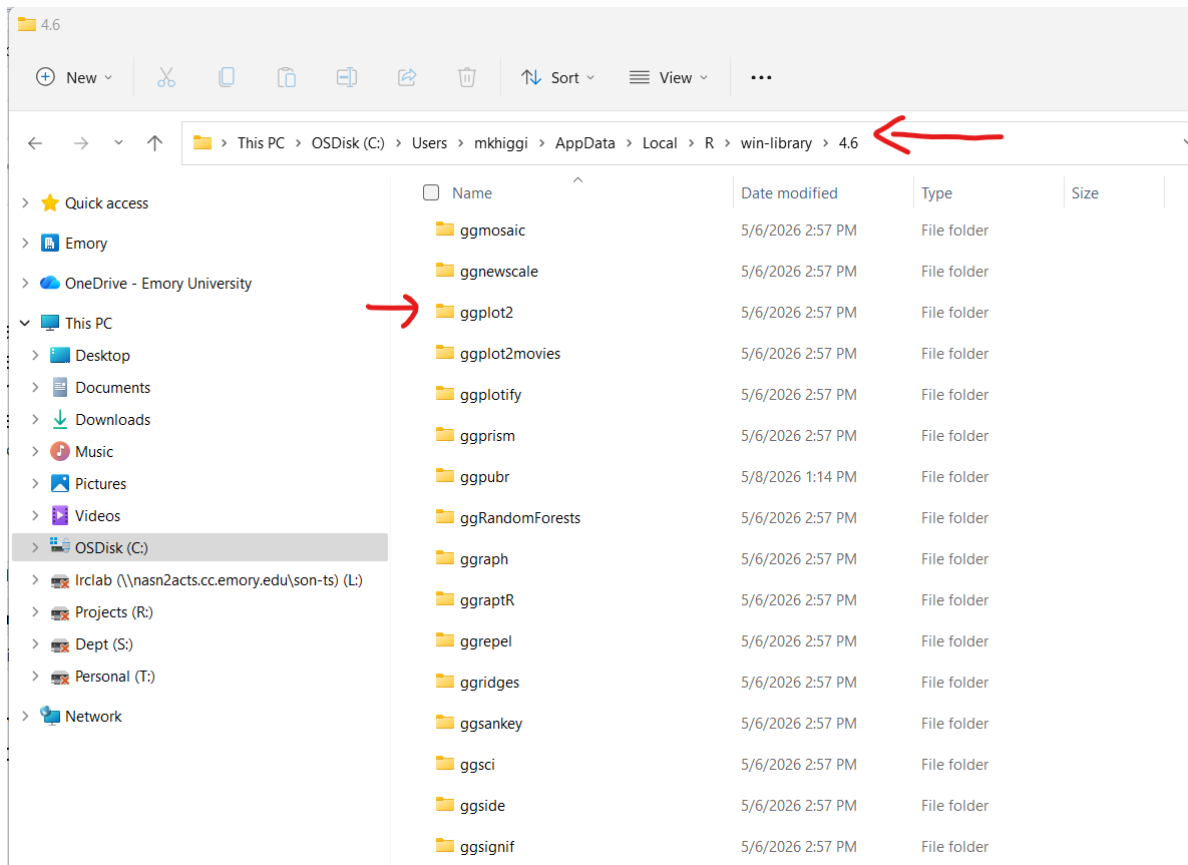
Install Cancel

After installing `ggplot2` you should see `ggplot2` listed under the Packages window:



Repeat this process to install the rest of the R packages needed.

Note: You can also confirm the installation of the R packages on your computer by going to the folder listed under #3 “Install to library” path listed above – on my computer my R packages are in this folder on my computer `C:\Users\mkhiggi\AppData\Local\R\win-library\4.6`.



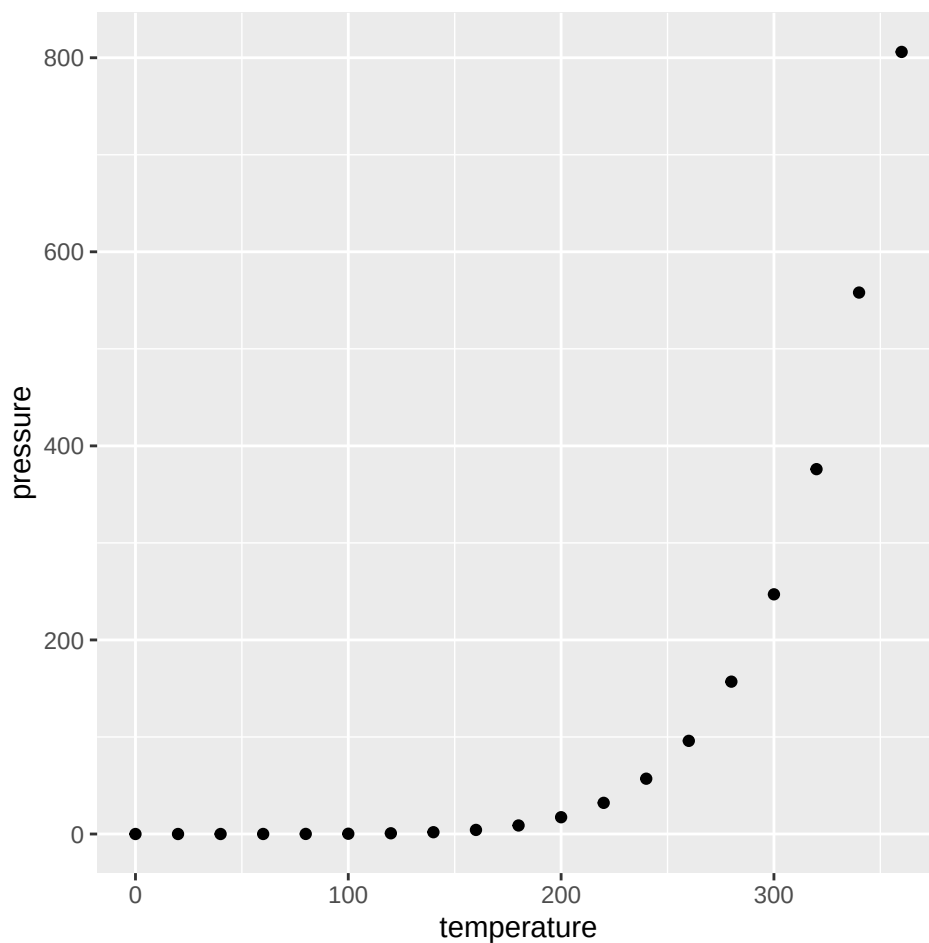
Learn More About Installing R Packages

- Chapter 1.3 in *Statistical Inference via Data Science: A ModernDive into R and the Tidyverse (Second Edition)*

4. Run Quick Check

Let's confirm that your installation of R, RStudio and your packages are working. For example, let's test that your installation of `ggplot2` is working. Run the following R code as a quick test.

```
library(ggplot2)
ggplot(pressure, aes(temperature, pressure)) +
  geom_point()
```

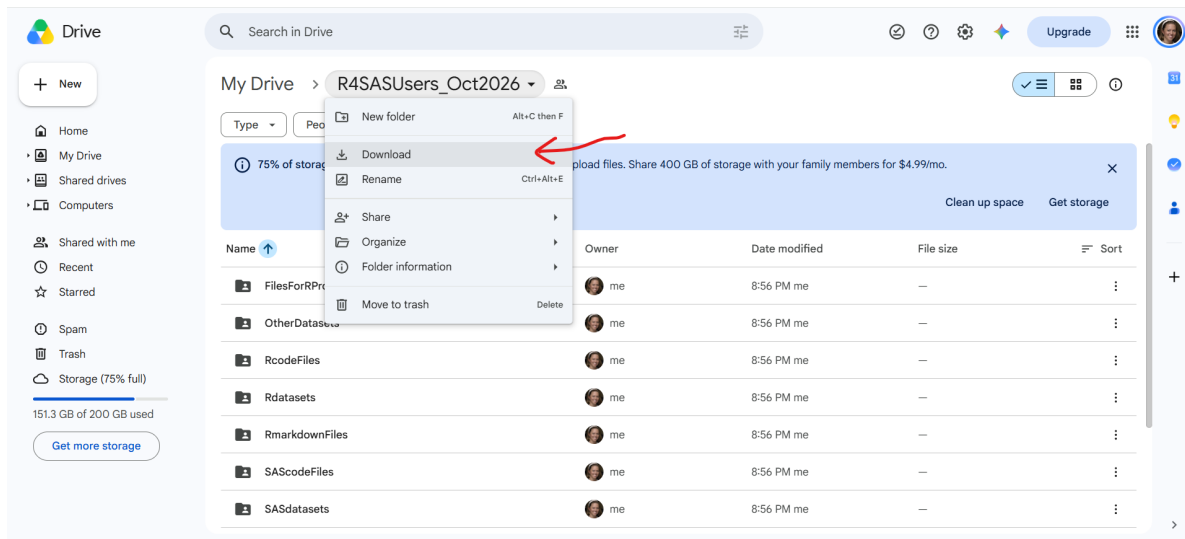


5. Download Files from Google Drive

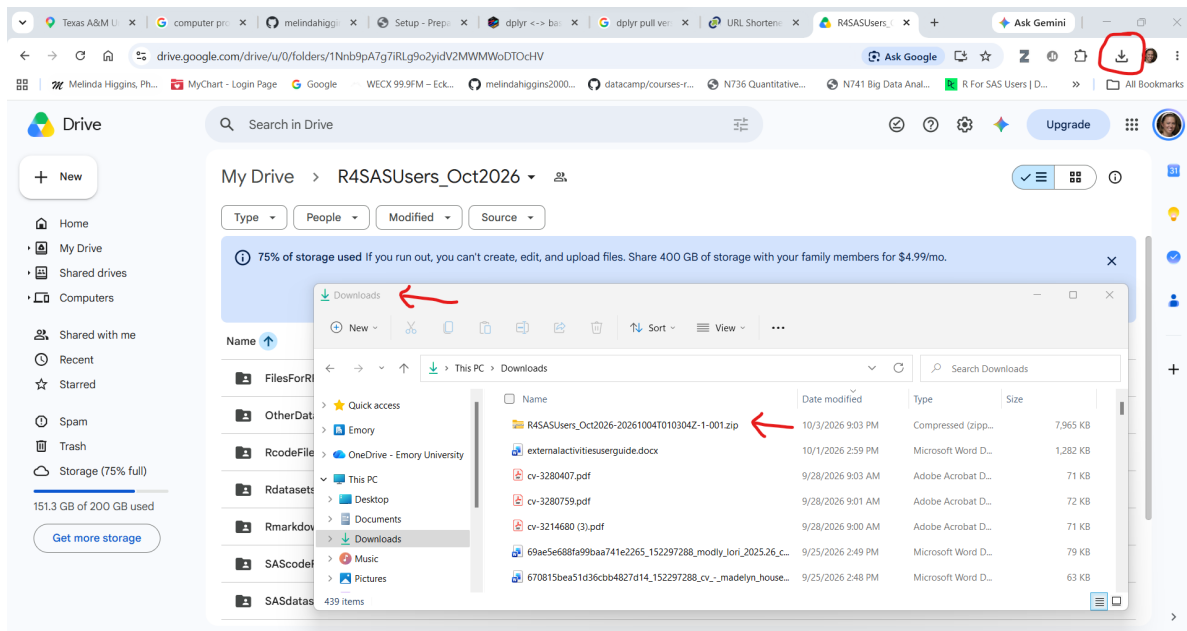
Now that you have R and RStudio ready to go, let's download the files you will need for this workshop.

First, go to the Google Drive [R4SASUsers_Oct2026](#) for this workshop.

Next, click the down arrow next to the folder name and select "Download".

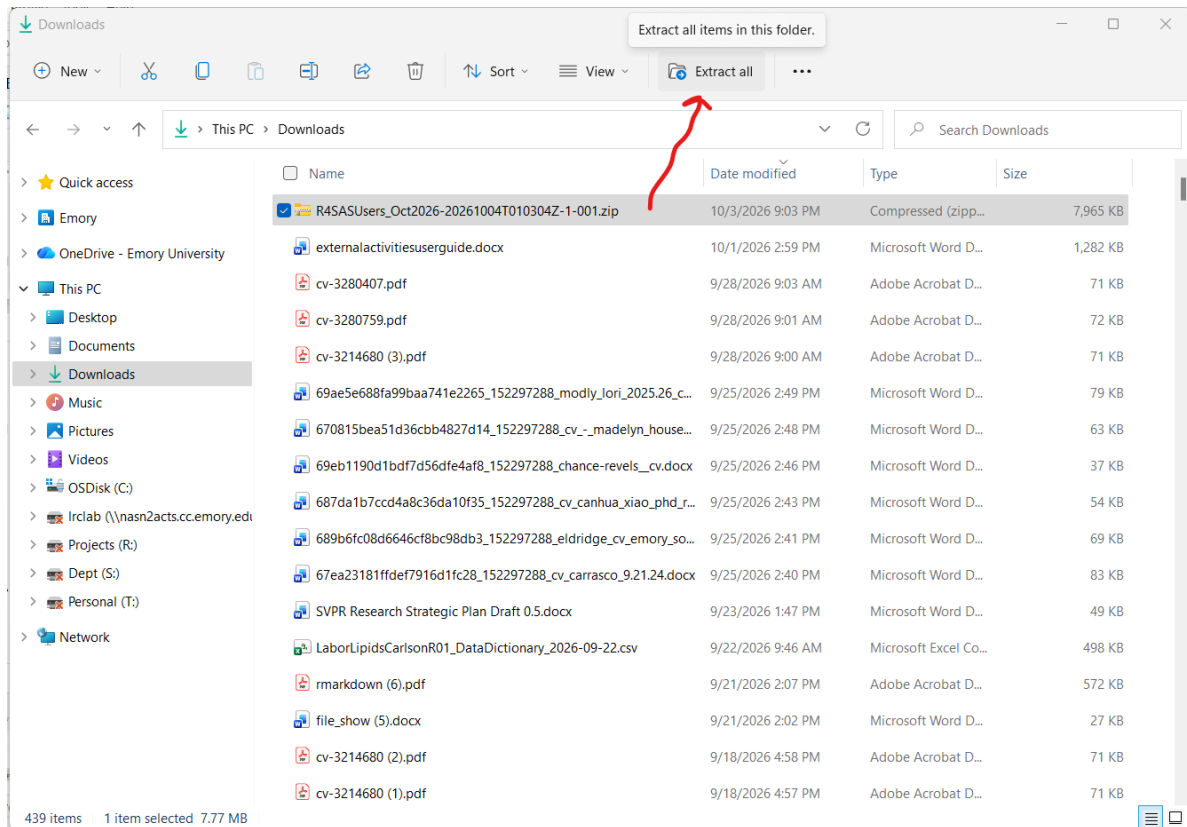


On my computer (Windows 11 operating system) using Google Chrome browser, this downloads a ZIPPED compressed file "R4SASUsers_Oct2026-20261004T010304Z-1-001.zip" - your actual filename may be different. But look for something similar.

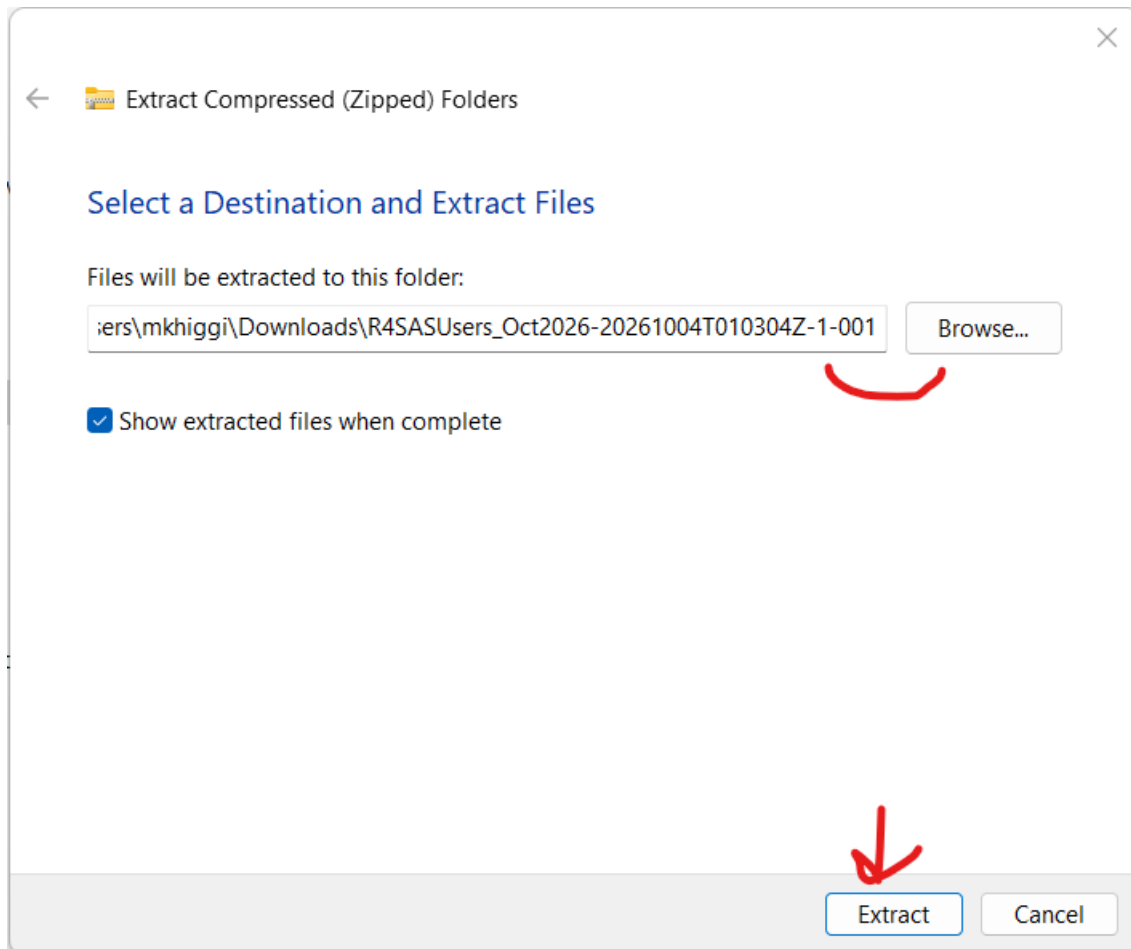


This *.ZIP file actually has all of the files that are on the Google Drive folder all in one compressed file. You have to UNZIP or EXTRACT these files before you can use them. The screenshots below are from a Windows 11 operating system - these may look different on different operating systems (e.g. Linux, Mac), but the process should be similar.

Click on “Extract All”

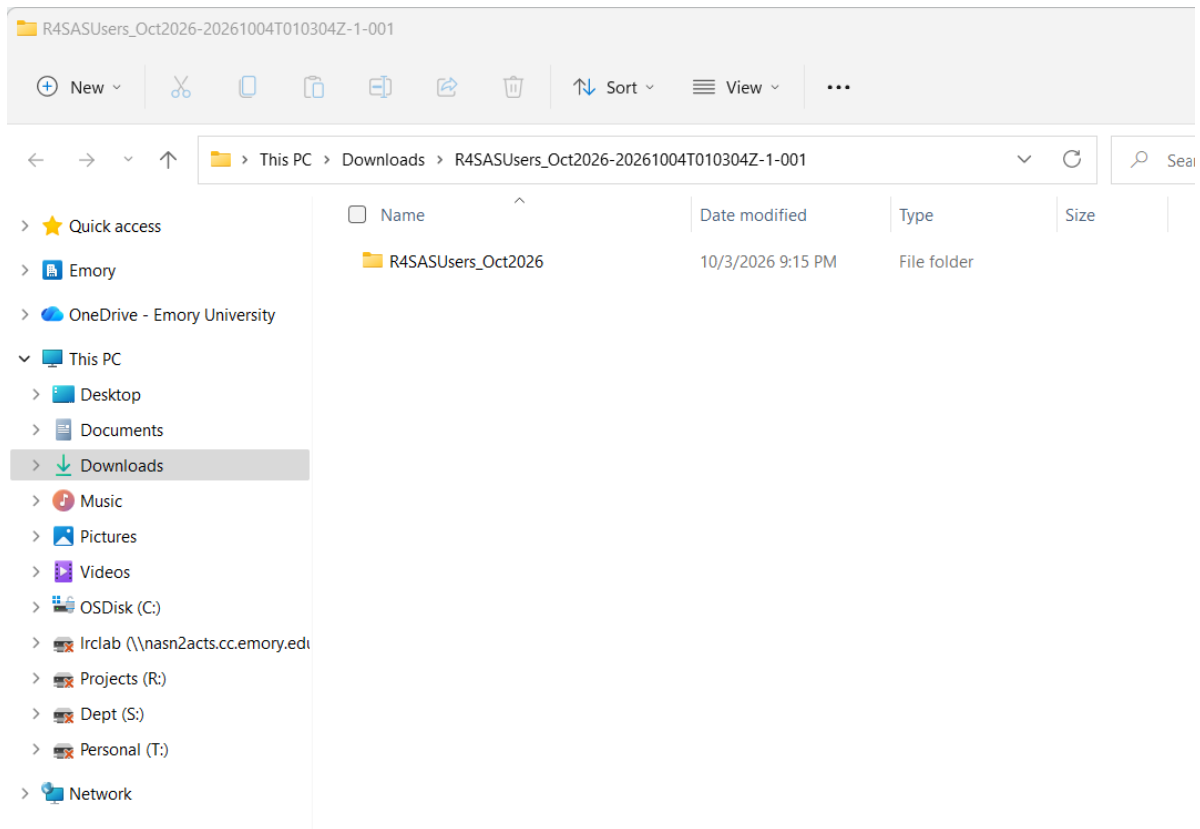


I usually accept the defaults for the folder location, but feel free to change this using “Browse” if you wish. Then click “Extract”.

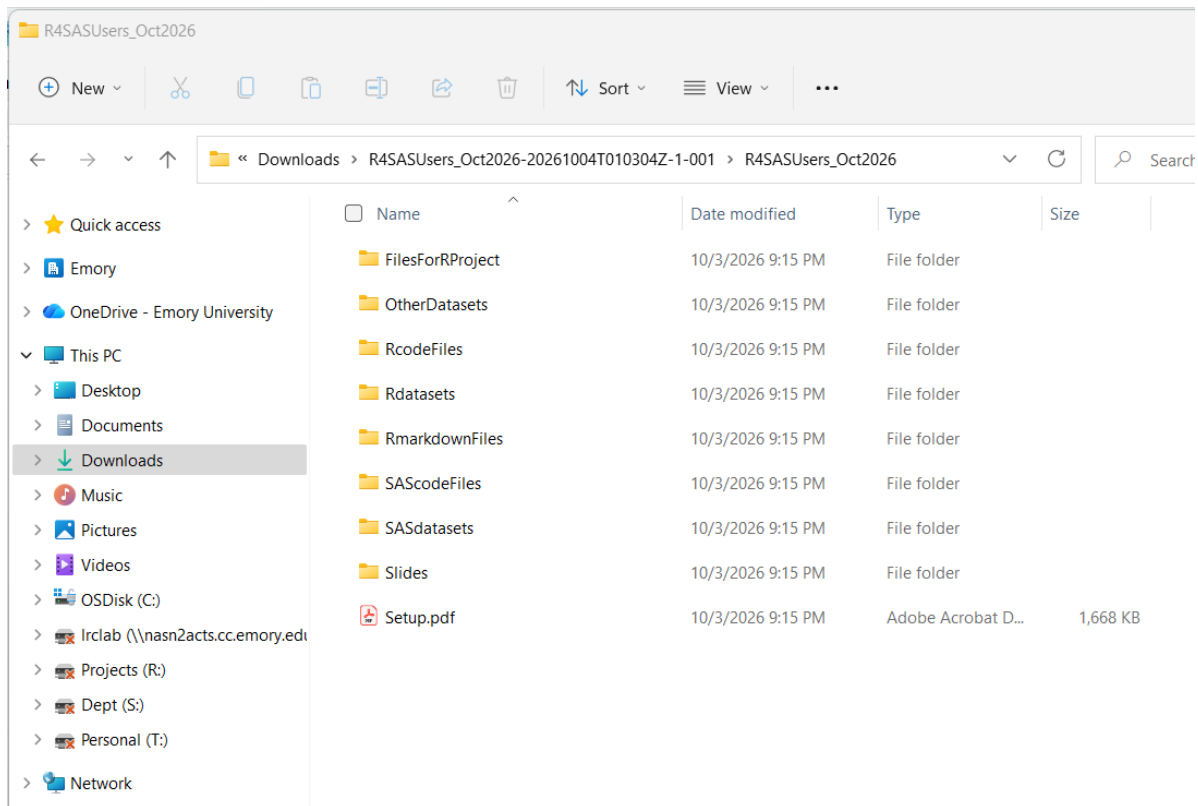


This then extracts all of the original files and places them into this new location on your computer (or cloud or network drive - your choice).

You should now have a new folder on your computer that looks like this:



Open this folder. These files and subfolders should now match what was on the Google Drive [R4SASUsers_Oct2026](#).



The files contained in this entire folder directory are:

Setup.pdf

```
FilesForRProject
  aggstats.R
  aggstats.sas
  CESDAnalysisReport.Rmd
  CESDAnalysisReport_byRace.Rmd
  CESDAnalysisReport_RaceParameters.Rmd
  d1_homework.R
  d1_homework_key.R
  d2_homework.R
  d2_homework_key.R
  dat1.Rdata
  dat1.sas7bdat
  dat2.Rdata
  dat2.sas7bdat
  finddups.R
  finddups.sas
```

```
help.csv
help.Rdata
help.sas7bdat
helpmkh.sav
letexample.R
letexample.sas
macroreg.R
macroreg.sas
module_d1_01.R
module_d1_02.R
module_d2_01.R
Rscript_01.R
sascode_d1_02.sas
sascode_d2_01.sas
saveplots.R
treat1.Rdata
treat1.sas7bdat
treat2.Rdata
treat2.sas7bdat
```

OtherDatasets

```
help.csv
helpmkh.sav
```

RcodeFiles

```
aggstats.R
d1_homework.R
d1_homework_key.R
d2_homework.R
d2_homework_key.R
finddups.R
letexample.R
macroreg.R
module_d1_01.R
module_d1_02.R
module_d2_01.R
Rscript_01.R
saveplots.R
```

Rdatasets

```
dat1.Rdata
dat2.Rdata
help.Rdata
```

```
treat1.Rdata
treat2.Rdata

RmarkdownFiles
  CESDAnalysisReport.Rmd
  CESDAnalysisReport_byRace.Rmd
  CESDAnalysisReport_RaceParameters.Rmd

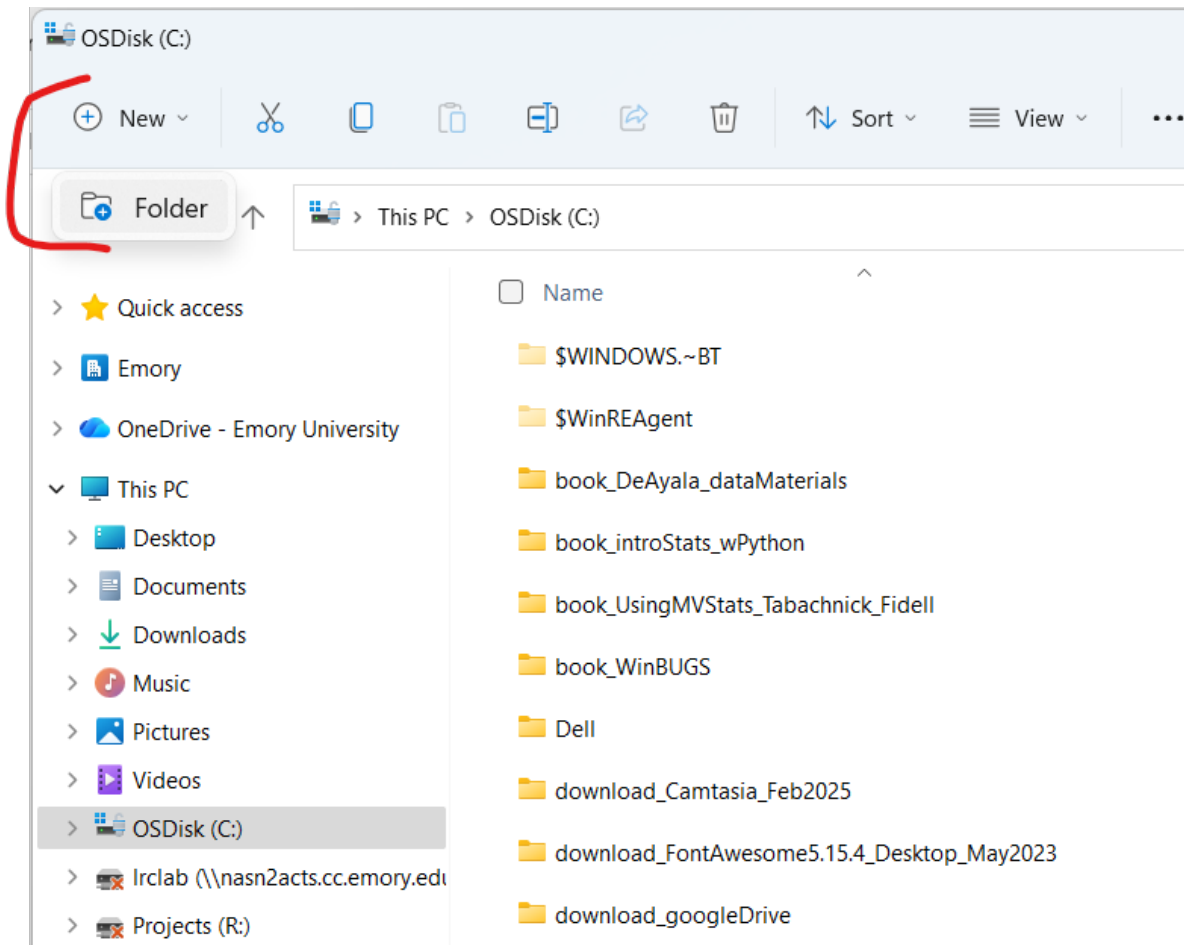
SAScodeFiles
  aggstats.sas
  finddups.sas
  letexample.sas
  macroreg.sas
  sascode_d1_02.sas
  sascode_d2_01.sas

SASdatasets
  dat1.sas7bdat
  dat2.sas7bdat
  help.sas7bdat
  treat1.sas7bdat
  treat2.sas7bdat

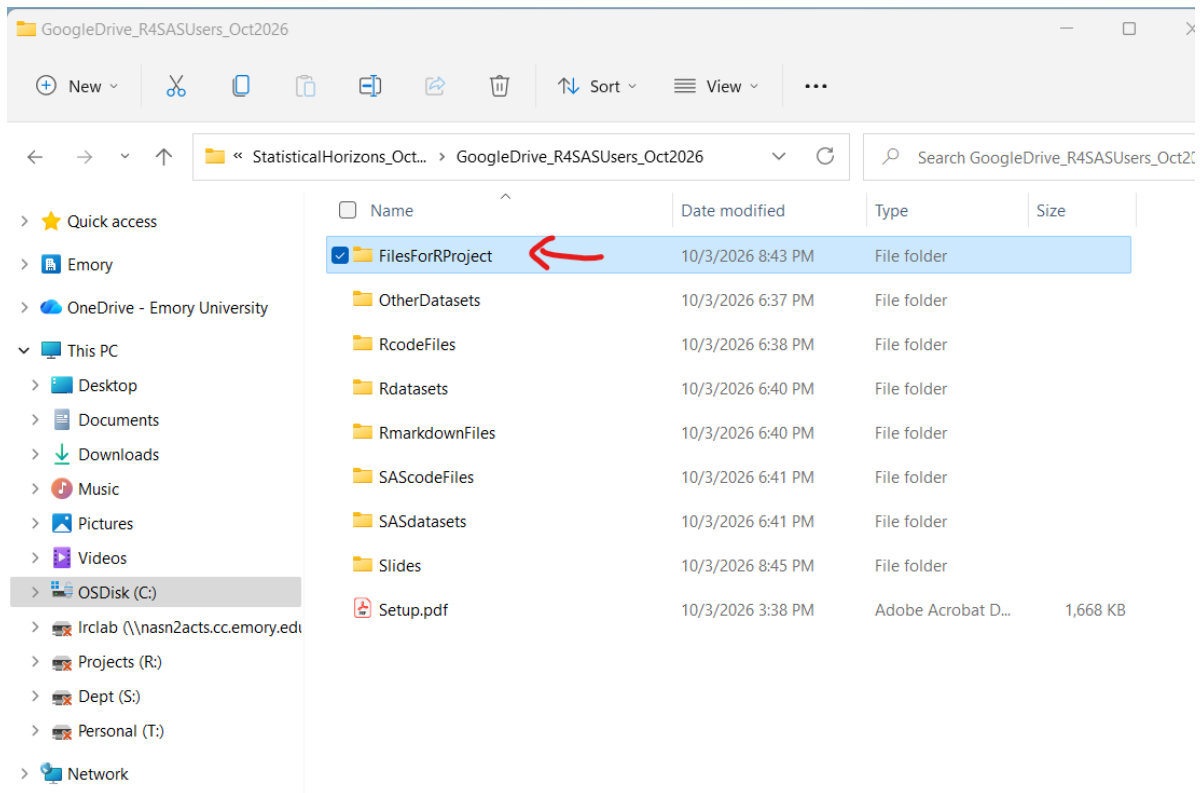
Slides
  RforSASUsers_d1_01.pdf
  RforSASUsers_d1_01.pptx
```

Next, we are going to move the files in the subfolder “FilesForRProject” into a new project folder on your computer.

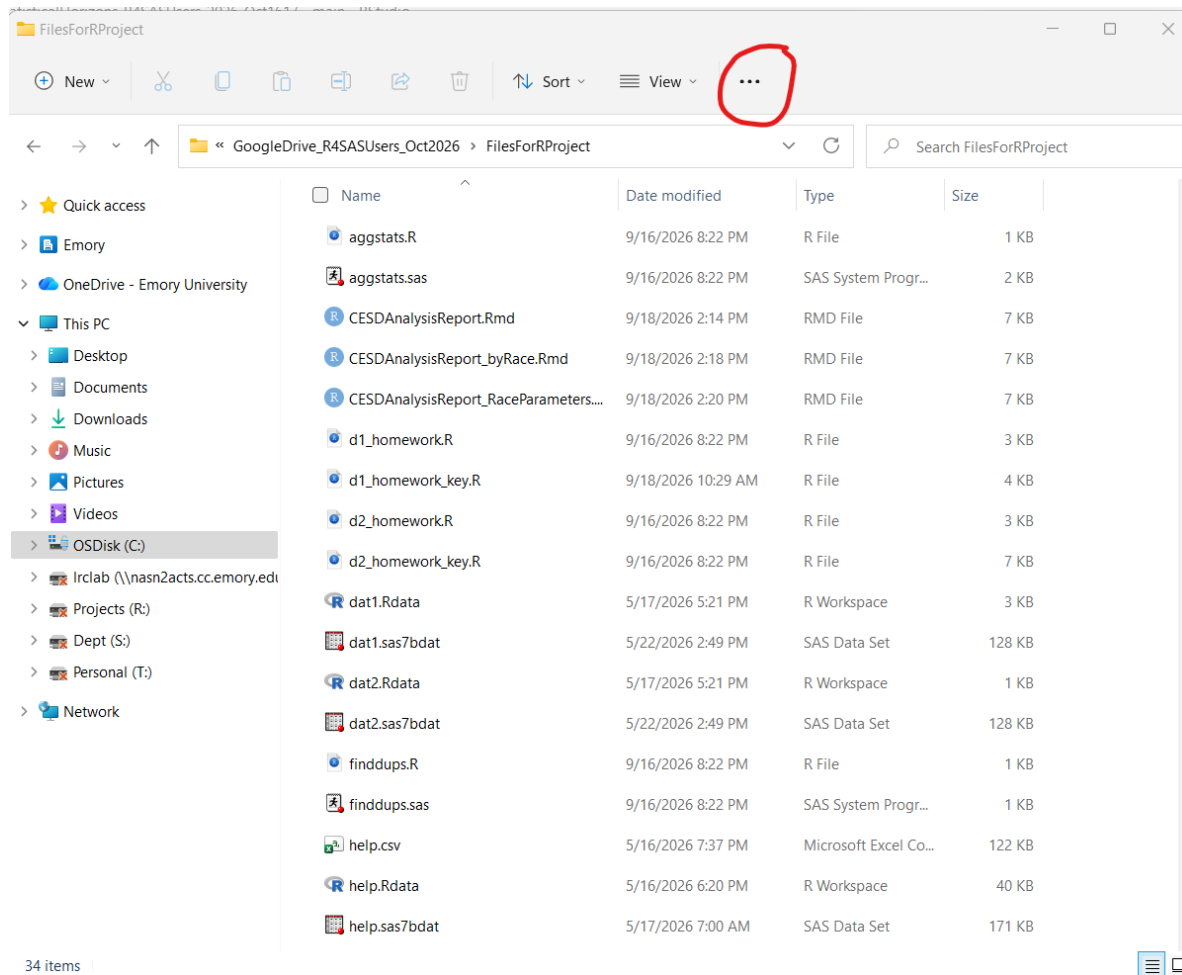
Create a new folder on your computer named “R4SASUsers”. Click on New/Folder and name this new folder “R4SASUsers”.

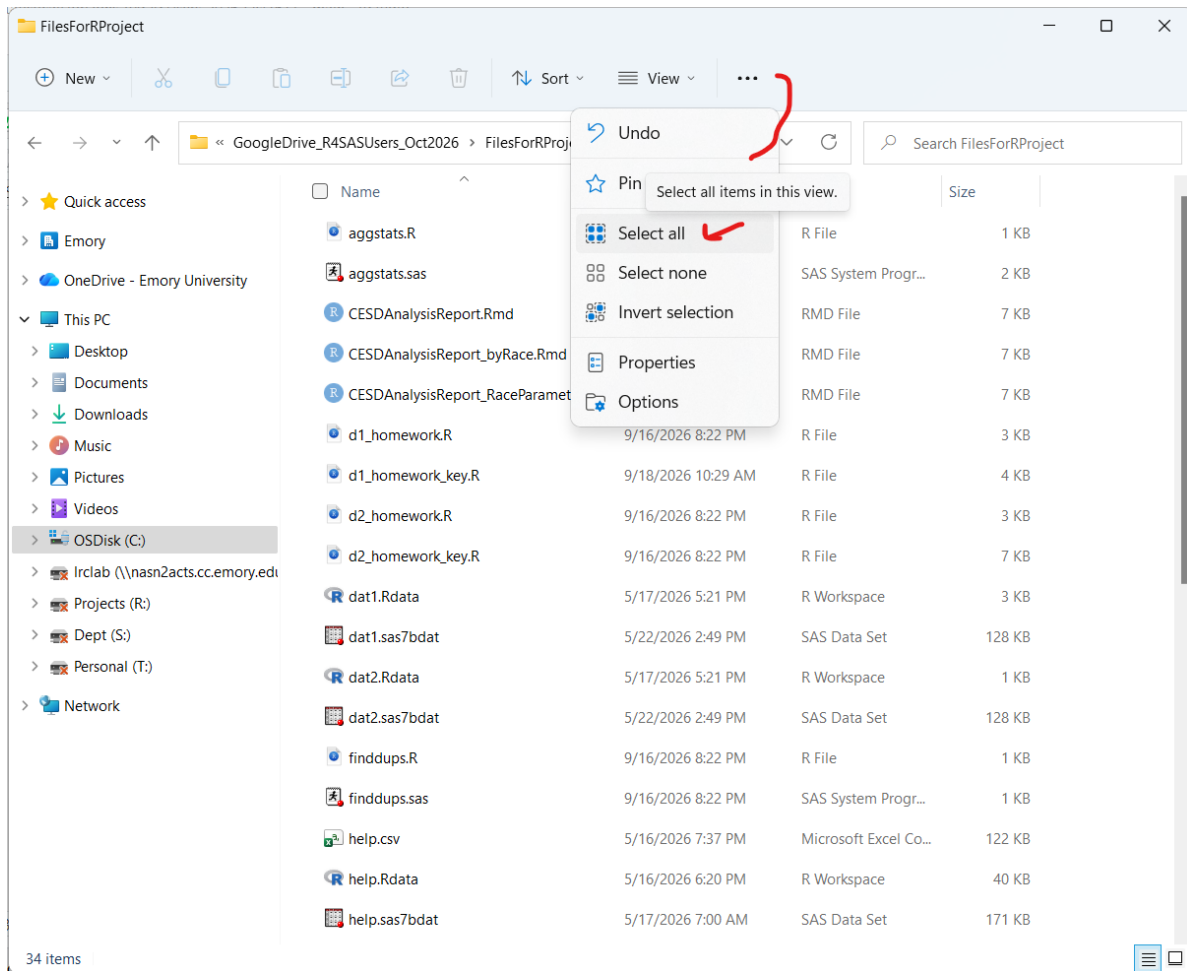


Open another file folder window and go back to where you unzipped the Google Drive files and folders. Then click on the subfolder “FilesForProject” to open it.

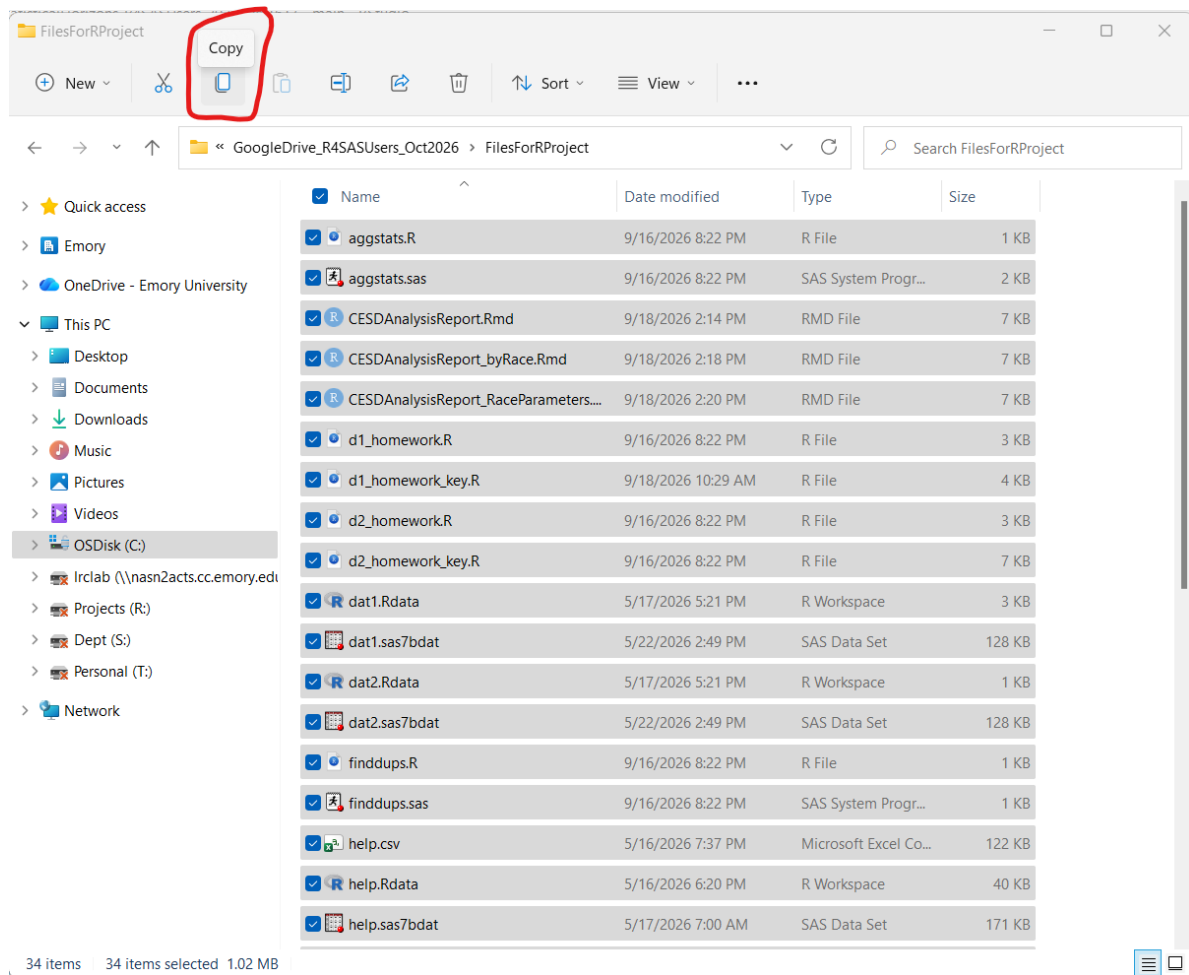


After opening the folder window, click on the 3 dots ... and choose “Select All”

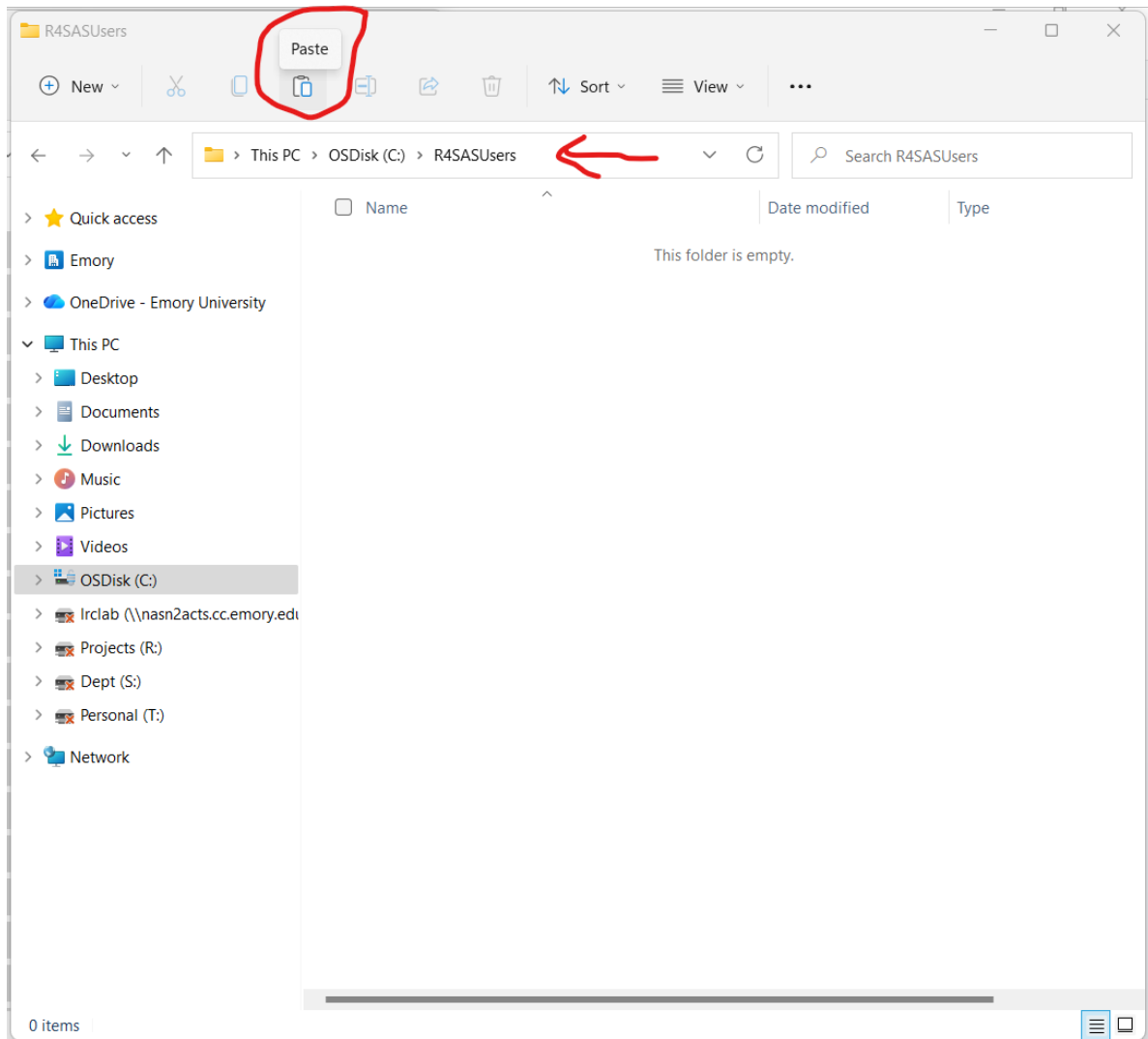


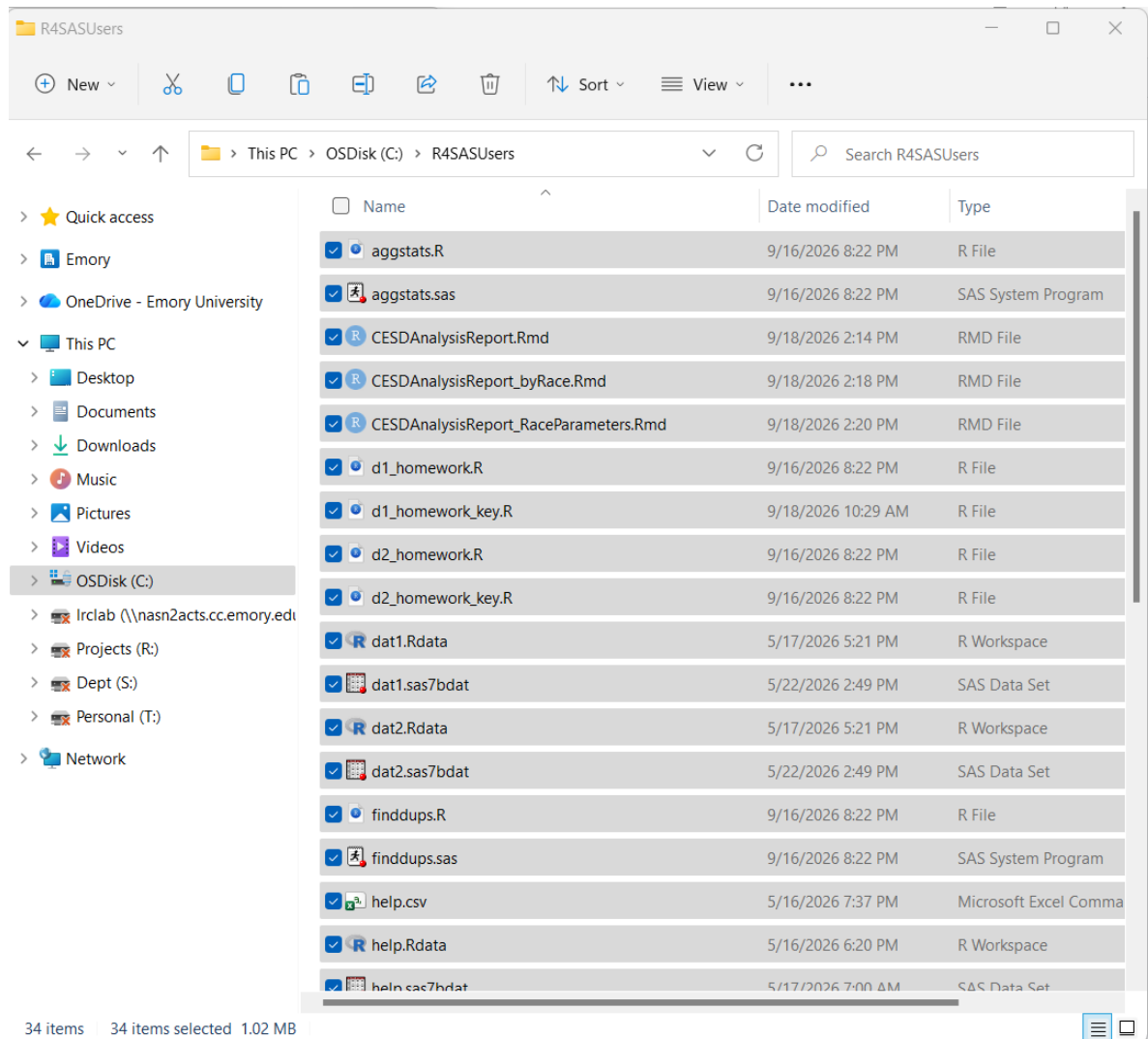


After selecting all of these files, choose COPY.



Then open the folder you just created and PASTE all of these files into your new folder.

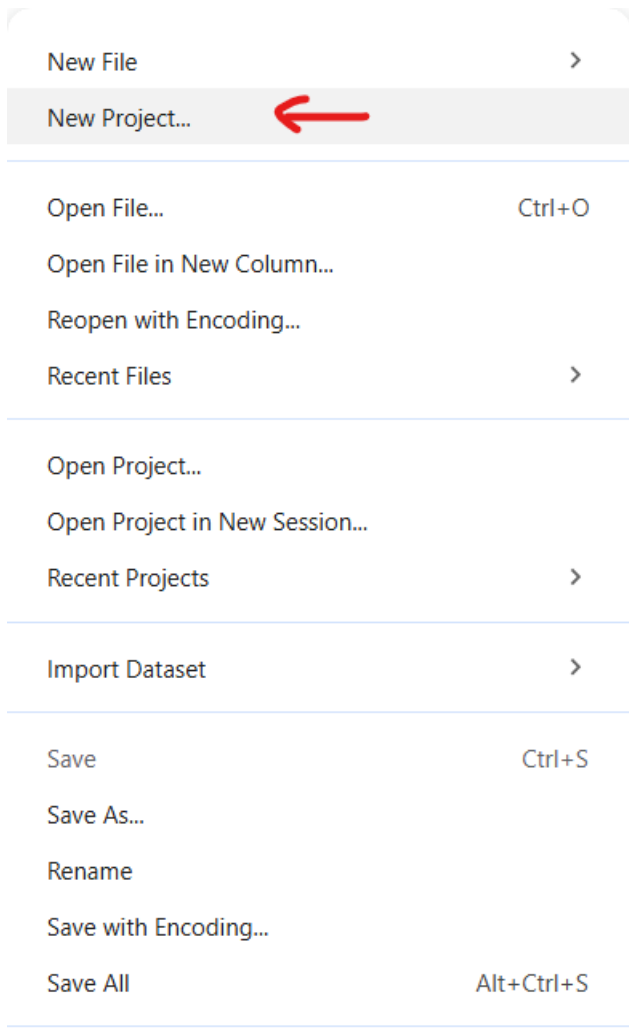




You now have your project folder created for this workshop. On my computer this folder location is `C:\R4SASUsers`.

6. Create New RStudio Project

Start RStudio and then Click on “File/New Project”.

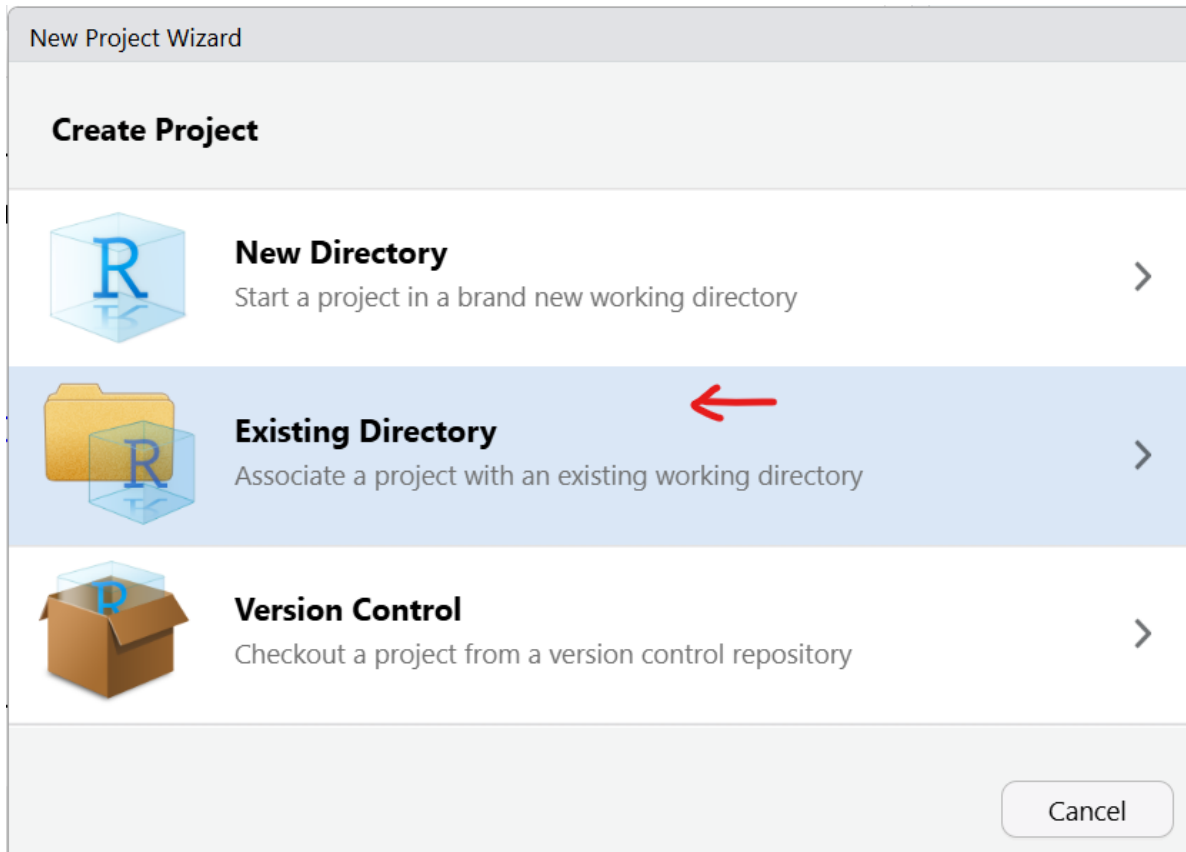


You have 3 options here:

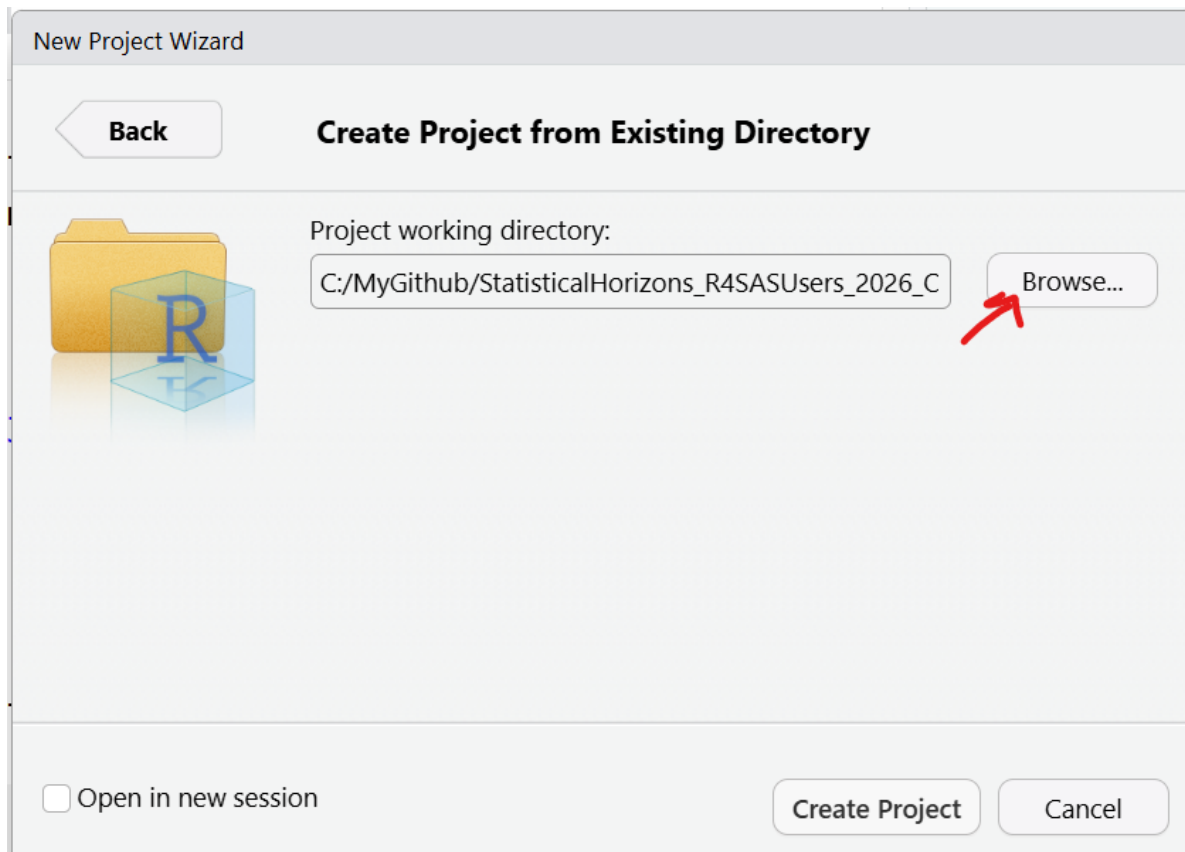
- New Directory
 - if the project folder does not exist yet
 - this is useful for more complicated “projects” like creating website or R packages
- Existing Directory
 - use this option if the folder where all of your project files are stored already exists
- Version Control

- use this option if you want to bring your project under version control using [Github](#) or [Subversion Control \(SVN\)](#).

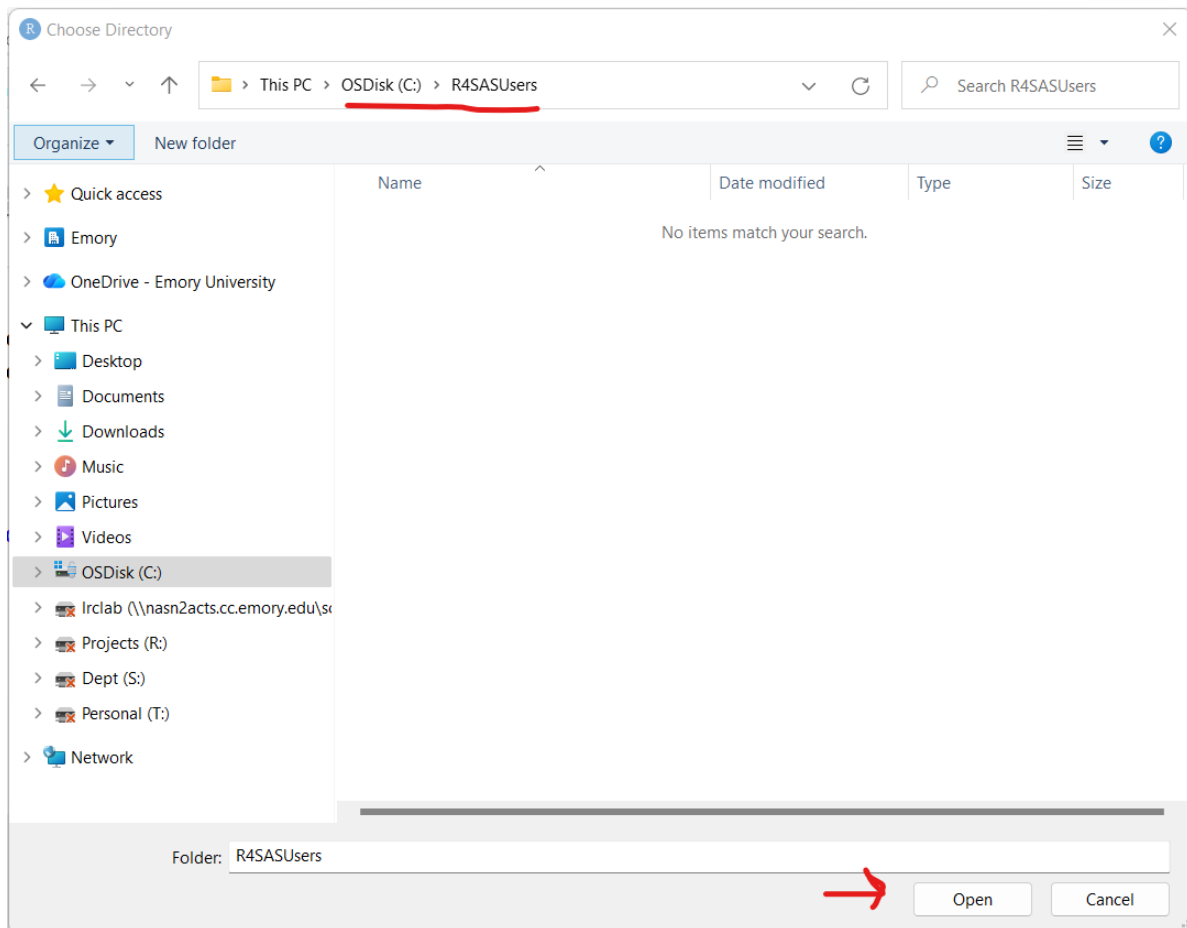
We will use “Existing Directory” since you have already created the project folder on your computer and put your files in there.



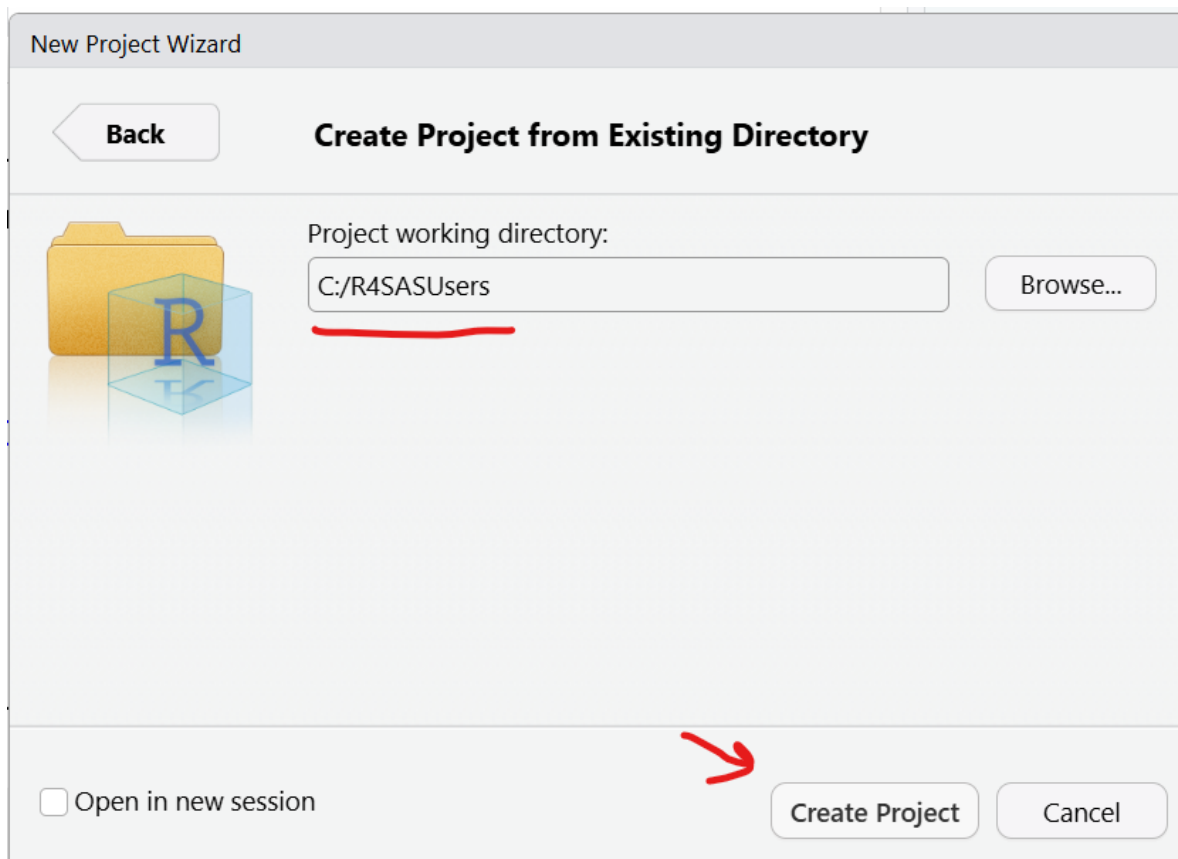
Next you need to select the existing project folder on your computer (or cloud or networked drive) - click “Browse”



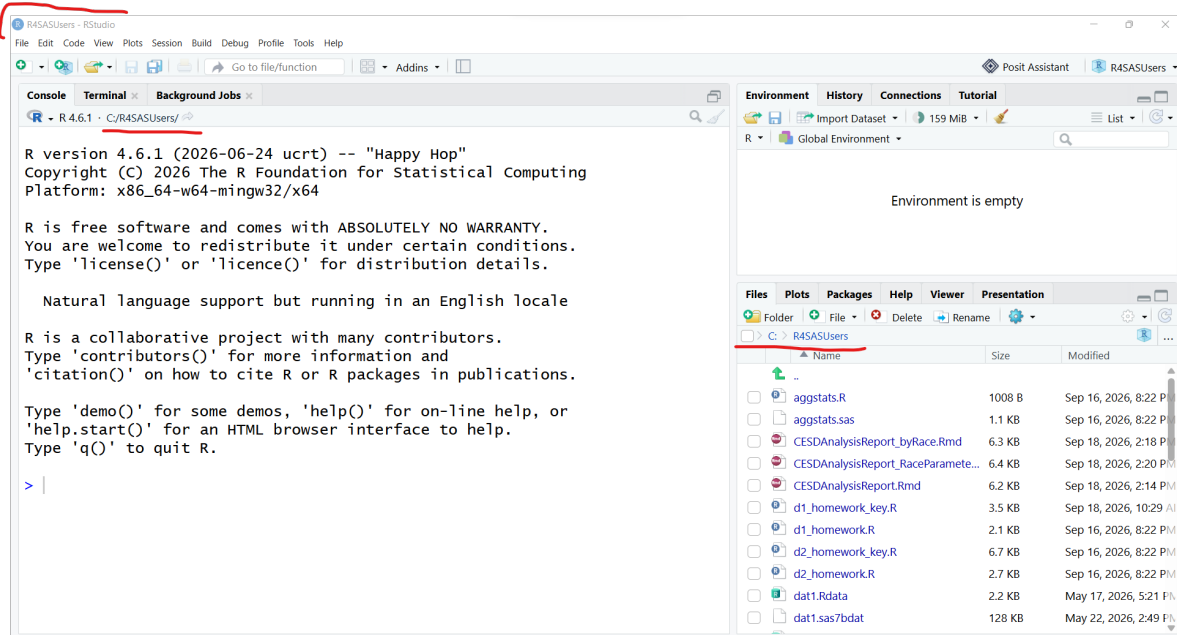
Then go find the folder and click “Open”.



Now check that your “Project working directory” matches the folder you selected - on my computer this is `C:\R4SASUsers`. Then click “Create Project” to create a new RStudio project.



Your new RStudio project should look like this. Check the folder location at the top left of your window. Also check that this is the SAME folder shown in the “Files” tab at the bottom right window. In this “Files” window, you should see the files you copied over from what you downloaded from the “FilesForRProject” folder from the Google Drive [R4SASUsers_Oct2026](https://tinyurl.com/r4sasusers) for this workshop.



CONGRATULATIONS You should be good to go!

References

- Grolemund, Garrett, and Hadley Wickham. 2011. “Dates and Times Made Easy with lubridate.” *Journal of Statistical Software* 40 (3): 1–25. <https://www.jstatsoft.org/v40/i03/>.
- Müller, Kirill, and Hadley Wickham. 2026. *Tibble: Simple Data Frames*. <https://tibble.tidyverse.org/>.
- R Core Team. 2026. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://doi.org/10.32614/R.manuals>.
- Spinu, Vitalie, Garrett Grolemund, and Hadley Wickham. 2023. *Lubridate: Make Dealing with Dates a Little Easier*. <https://lubridate.tidyverse.org>.
- Wickham, Hadley. 2016. *Ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York. <https://ggplot2.tidyverse.org>.
- Wickham, Hadley. 2023. *Tidyverse: Easily Install and Load the Tidyverse*. <https://tidyverse.tidyverse.org>.
- Wickham, Hadley. 2025a. *Forcats: Tools for Working with Categorical Variables (Factors)*. <https://forcats.tidyverse.org/>.
- Wickham, Hadley. 2025b. *Stringr: Simple, Consistent Wrappers for Common String Operations*. <https://stringr.tidyverse.org>.
- Wickham, Hadley, Mara Averick, Jennifer Bryan, et al. 2019. “Welcome to the tidyverse.” *Journal of Open Source Software* 4 (43): 1686. <https://doi.org/10.21105/joss.01686>.
- Wickham, Hadley, Winston Chang, Lionel Henry, et al. 2025. *Ggplot2: Create Elegant Data Visualisations Using the Grammar of Graphics*. <https://ggplot2.tidyverse.org>.
- Wickham, Hadley, Romain François, Lionel Henry, Kirill Müller, and Davis Vaughan. 2026. *Dplyr: A Grammar of Data Manipulation*. <https://dplyr.tidyverse.org>.
- Wickham, Hadley, and Lionel Henry. 2025. *Purrr: Functional Programming Tools*. <https://purrr.tidyverse.org/>.
- Wickham, Hadley, Jim Hester, and Jennifer Bryan. 2026. *Readr: Read Rectangular Text Data*. <https://readr.tidyverse.org>.
- Wickham, Hadley, Davis Vaughan, and Maximilian Girlich. 2025. *Tidyr: Tidy Messy Data*.

<https://tidyr.tidyverse.org>.

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